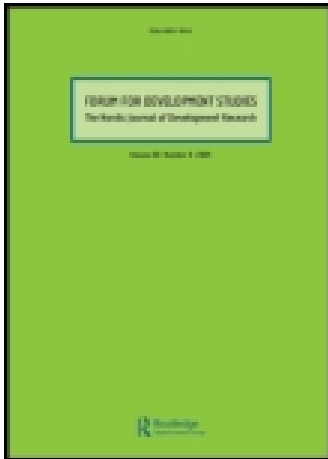




Forum for Development Studies

Publication details, including instructions for authors and subscription information:

<http://www.tandfonline.com/loi/sfds20>

Back to the Barriers? Changing Narratives in Biodiversity Conservation

Jon Hutton^a, William M. Adams^b & James C. Murombedzi^c

^a Resource Africa

^b University of Cambridge

^c World Conservation Union

Published online: 28 Jan 2011.

To cite this article: Jon Hutton, William M. Adams & James C. Murombedzi (2005) Back to the Barriers? Changing Narratives in Biodiversity Conservation, Forum for Development Studies, 32:2, 341-370, DOI: [10.1080/08039410.2005.9666319](https://doi.org/10.1080/08039410.2005.9666319)

To link to this article: <http://dx.doi.org/10.1080/08039410.2005.9666319>

PLEASE SCROLL DOWN FOR ARTICLE

Taylor & Francis makes every effort to ensure the accuracy of all the information (the "Content") contained in the publications on our platform. However, Taylor & Francis, our agents, and our licensors make no representations or warranties whatsoever as to the accuracy, completeness, or suitability for any purpose of the Content. Any opinions and views expressed in this publication are the opinions and views of the authors, and are not the views of or endorsed by Taylor & Francis. The accuracy of the Content should not be relied upon and should be independently verified with primary sources of information. Taylor and Francis shall not be liable for any losses, actions, claims, proceedings, demands, costs, expenses, damages, and other liabilities whatsoever or howsoever caused arising directly or indirectly in connection with, in relation to or arising out of the use of the Content.

This article may be used for research, teaching, and private study purposes. Any substantial or systematic reproduction, redistribution, reselling, loan, sub-licensing, systematic supply, or distribution in any form to anyone is expressly forbidden.

Back to the Barriers? Changing Narratives in Biodiversity Conservation

Jon Hutton
Director,
ResourceAfrica

William M. Adams
Professor,
University of Cambridge

James C. Murombedzi
Regional Director,
World Conservation
Union

Abstract

The dominant approach to conservation in the 20th century was the establishment of protected areas from which people were excluded. However, in the 1980s, decentralised, community-based approaches to biodiversity conservation and natural resource management began to spread rapidly, especially in southern Africa. From the early 1990s, there has been a growing divide between proponents of community-based approaches to conservation (particularly community-based natural resource management, CBNRM) and those advocating a return to more traditional preservationist approaches to biodiversity conservation. Here we examine the growth of the community narrative and the subsequent revival of what we call the 'back to the barriers' movement. We discuss the importance of various actors and sets of policy ideas to this revival in Africa. Changes in narratives have had profound impacts upon conservation and natural resource management, livelihood strategies and political processes. We suggest that policy debate needs to become less formulaic if outcomes are to be positive.

Keywords: biodiversity conservation, community-based natural resource management, community conservation, narratives, protected areas, Global Environment Facility, Convention on Biological Diversity, trans-boundary natural resource management, direct payments, cross-sector partnerships, mainstreaming, poverty alleviation

1. Introduction

Historically, conservation strategies have been dominated by attempts to reserve places for nature, and to separate humans and other species (Grove, 1995; MacKenzie, 1988; Adams, 2003,

Note of acknowledgements. We are grateful to many colleagues for their stimulating discussion around the issues we have covered in this article, notably colleagues at the Programme for Land and Agrarian Studies at the University of the Western Cape in South Africa, and the UK Poverty and Conservation Working Group. We would also like to acknowledge with thanks the important contributions of Kule Chitepo and Barney Dickson of ResourceAfrica and Dilys Roe of IIED.

2004). The model of conservation came to dominate 20th century thinking, drawing in particular on the US idea of a national park as a pristine or wilderness area, and the British notion of an intensively managed nature reserve. This has been variously (and often pejoratively) called 'fortress conservation' or the 'fences and fines approach' (Wells and Brandon, 1992; Neumann, 1998; Brockington, 2002). Such widely shared ideas about policy are referred to as 'narratives' (Roe, 1991; Leach and Mearns, 1996), and that term is used in this article. Fortress conservation involved the creation of protected areas, the exclusion of people as residents, the prevention of consumptive use and minimisation of other forms of human impact (Brockington and Schmidt-Soltau, 2004). This narrative was very influential in sub-Saharan Africa, and there is a long history of national park creation from the colonial game reserves established in many newly-annexed territories in the 1890s to the Parc National Albert established in the Belgian Congo in 1925 and the Kruger National Park in Natal in 1926, through to the flood of national parks created in the decades between the end of the second world war and the widespread end of colonial rule in the 1960s, such as Nairobi National Park (1946), Tsavo (1948), and Serengeti and Murchinson Falls (1951) (Carruthers, 1995; Adams, 2004; Neumann, 1998).

By the 1990s, the dominant narrative of fortress conservation no longer enjoyed hegemony, either in Africa or globally. It had progressively been challenged by a new community conservation narrative which stressed the need not to exclude local people, either physically from protected areas or politically from the conservation policy process, but to ensure their participation (Western *et al.*, 1994; Adams and Hulme, 2001). The new 'community conservation' initiatives were diverse, including community-based conservation, community wildlife management, collaborative or co-management, community-based natural resource management, state/community co-management and integrated conservation and development programmes (Barrow and Murphree, 2001).

This article draws on experience in southern Africa to examine the growth of the community narrative and the subsequent revival of the fortress conservation narrative. This revival is significant in scale and scope, and we give it a title, the 'back to the barriers' movement. We extend the arguments of Wilshusen *et al.* (2002) and Brechin *et al.* (2003) to analyse arguments that are made against community approaches to conservation and in favour of a return to an exclusive, protectionist approach. We also note that the success of proponents of the 'back to the barriers' narrative has partly been

because they identify real shortcomings in community approaches. However, we suggest that their own arguments also contain false assumptions and misplaced arguments. We explore concerns that, without further guidance, some policy-makers might conclude that ideas such as sustainable use, community-based conservation and co-management are so deeply flawed that they have no role in biodiversity conservation, natural resource management and, most especially, protected area management. We outline some of the actors supporting the 'back to the barriers' revival, and describe some of the policy ideas that have been factors in the movement's renewed ascendancy (such as trans-boundary natural resource management, 'direct payments', public--private partnerships and mainstreaming). We argue that these, paradoxically, pose significant challenges to effective conservation in many parts of the developing world.

2. The Rise of Community-based Approaches to Conservation

The negative impacts on local people of areas set aside to protect biodiversity have been widely discussed (e.g. Brockington, 2002; Brockington and Schmidt-Soltau, 2004, Colchester, 1997, 1998, 2003; Brechin *et al.*, 2003; Chapin, 2004) and it is tempting to suggest that the narrative of fortress conservation was challenged largely in recognition of its human costs, brought to general attention in the African context by Adams and McShane (1992). However, our personal experience suggests that change largely resulted from the self-interest of the conservation constituency, which, during the late 1970s, correctly recognised that fortress conservation would be difficult to maintain politically in the face of objections by local people and their political leaders, in countries with renewed democracies. The need for a 'community' approach to protected areas was articulated at successive World Congresses on National Parks and Protected Areas, particularly the third in 1982 and the fourth in 1992 (McNeely and Miller, 1984; McNeely, 1992; Kemf, 1993). It was a key element in the concept of biosphere reserves developed in the 1970s by the Man and the Biosphere (*sic*) programme, and was recognised in the WWF's Wildlife and Human Needs Programme (1985), and in a plethora of 'people and park' projects developed in the late 1980s worldwide (Hannah, 1992). The Fifth World Parks Congress in 2003 had as its theme 'benefits beyond boundaries' (Steiner, 2003).

Conservation began to be linked to local development needs

through such initiatives as ‘conservation-with-development projects’ (Stocking and Perkin, 1992), community-based natural resource management (Fabricius *et al.*, 2004) or ‘integrated conservation and development projects’ (ICDPs) (Wells and Brandon, 1992; Barrett and Arcese, 1995). The community conservation narrative, in contrast to the then-dominant protected area narrative, recognised both the moral implications of costs of conservation borne by local people, and the pragmatic problem of the hostility of displaced or disadvantaged local people to conservation organisations practising a ‘fortress conservation’ policy (Adams and Hulme, 2001; Wells and Brandon, 1992; Western *et al.*, 1994).

Initially, the shift away from the ‘fortress’ approach towards a broader vision of conservation within an inhabited economic landscape, was widely endorsed by supporters of the idea of protected areas, who recognised that protected areas in biodiversity-rich developing countries were managed ineffectively, if at all, and thus provided little or no protection for biodiversity (Brandon *et al.*, 1998; Bruner *et al.*, 2001). Conservation scientists also started to point out that protected areas were usually too small to meet the requirements of either ecosystem or biodiversity conservation. Research in island biogeography, translated to terrestrial habitat fragments, suggested that biodiversity could not be sustained on small preserved habitat ‘islands’ (see, for example, Shafer 1990). Even if conservation ‘fortresses’ could be established, paid for and policed, they would not be enough. Conservation had to reach out of protected areas to the human communities of the increasingly densely inhabited wider landscape.

The rapid acceptance of the community conservation narrative within international policy was due to at least four reasons. First, it tied conservation to sustainable development, and captured the upwelling of political and policy commitment to sustainable development arising from the Brundtland Report (1987) and the UN Conference on Environment and Development, held in Rio in 1992. Second, it drew on the rediscovery of idealist and romantic ideas about the ‘community’ (and ‘local’ and ‘grass-roots’ action, and the participatory approach) as an alternative to the state as a means of achieving positive social change. Communitarian development became popular in the West during the 1980s, notably in the ‘communitarian’ movement in the USA (Adams and Hulme, 2001). Third, the community conservation narrative also benefited from ideas crossing over from the development arena. It emerged at a time of significant shifts in the dominant discourses of develop-

ment from the 'top-down', 'technocratic' models of the 1970s to 'bottom-up', 'decentralised', and 'participatory' planning (Turner and Hulme, 1997). The fourth reason for the success of the community conservation narrative was the way it fitted with the renewed interest in the 1980s in the market as an alternative to the state as a means of delivering policy change. The market was a central feature of the 'New Policy Agenda' for foreign aid that was developed in particular by the USA in the early 1990s. To achieve public policy goals (including conservation) economic incentives needed to be set correctly for all actors, and this was best done by market mechanisms and not by state planning. Communities, and rural individuals and households, should become micro-entrepreneurs, using the economic values of conservation resources (for tourism, trophy-hunting, medicines, meat or other products) to deliver both sustainable livelihoods and conservation.

Community approaches to conservation in the 1980s and 1990s can be thought of as a continuum from protected-area outreach to programmes building on local capacity to manage resources sustainably (Barrow and Murphree, 2001). Within southern Africa, community-based natural resource management (CBNRM) was most important in the 1980s and 1990s (Fabricius *et al.*, 2004). This co-evolved in several different contexts in response to a range of historical, political, social and economic experiences, conditions and challenges. In southern Africa, attempts at engaging communities in natural resources management (NRM) were informed by the idea of extending to residents of communal lands some benefits from resource use and management as enjoyed by landowners on leasehold and title-hold land, as in Zimbabwe, or by a need to respond to the challenges of engaging traditional authorities in the management of the benefits of hunting in state 'game management areas', as in Zambia. These early programmes influenced the development and growth of a suite of programmes for the management of natural resources elsewhere in Africa and, through the international attention paid to programmes such as Zimbabwe's CAMPFIRE, more widely.

The early CBNRM programmes focused almost exclusively on wildlife management, in the form of safari hunting. This was largely because the early amendments to legislation were made in the wildlife sector, and because wildlife use produced significantly higher and more visible benefits than other resources, while the maintenance of wildlife within a partly farmed landscape also incurred more costs, and therefore conflicts, between wildlife and people (Hughes, 2001). Subsequent developments in policy and

legislation have made it possible for communities to gain some control over forests and timber resources, as well as other non-timber forest products such as grasses or medicines, and activities such as grazing. Today, all countries in the region have developed policies and enacted laws that allow communities in communal areas to use and benefit from the natural resources of their lands, with varying degrees of success (Fabricius *et al.*, 2004).

CBNRM programmes developed on the back of several premises, namely that:

- ▼ communities are more efficient managers of natural resources in their areas of jurisdiction than other agencies;
- ▼ community management leads to improved incomes for communities, which both plays a significant role in poverty reduction and provides economic incentives for conservation;
- ▼ community management reduces conflicts with wild animals, and thus the cost they impose on people, leading to better tolerance of wildlife and better outcomes for biodiversity;
- ▼ the community management of natural resources is more efficient than state management, improves efficiency and reduces the costs of management.

Very quickly, within perhaps five years, conservation practitioners in southern Africa began to style themselves as working for conservation ‘by the people’, ‘with the people’ or ‘for the people’, rather than ‘against the people’, as in fortress conservation (Child, 2004).

3. Back to the Barriers

While conservation planners in IUCN and elsewhere, spurred on by the indigenous peoples’ movement and debate at successive World Parks Congresses and world summits on sustainable development, sought to entrench approaches to conservation that built in the involvement of local people, the alternative ‘fortress conservation’ strategy began to gain new salience among conservation biologists. The case for strict protection is widely made, and its elements were marked out in a number of seminal papers during the early 1990s. Thus Oates (1995) argued that only strictly protected areas offered a chance of preserving African forest primates, and Redford (1991, 1992) argued that human presence in tropical forests was ultimately incompatible with the conservation of biological diversity. The case for strictly protected parks was made in a number of influential

books that appeared in the late 1990s such as *Requiem for Nature* (Terborgh, 1999), *Myth and Reality in the Rain Forest* (Oates, 1999), *Last Stand: Protected areas and the defense of tropical biodiversity* (Kramer *et al.*, 1997) and *Parks in Peril: People, Politics and Protected Areas* (Brandon *et al.*, 1998). This 'resurgence' of the 'protectionist paradigm' (Wilshusen *et al.*, 2002) was in no sense accidental, but reflected a strong US-led vision of conservation. This stressed the great urgency of conservation action (see, for example, Wilson, 1992), the need for science-based conservation planning (Margules and Pressey, 2000) and the completion of global scientific analysis of areas of greatest biodiversity, such as the 'hotspots' concept adopted by Conservation International (Myers *et al.*, 2000). There was a parallel development in the scale and power of conservation organisations (Brosius, 1999).

A series of arguments are commonly put forward in support of the protectionist case. Wilshusen *et al.* (2002) identify five core elements of the 'protectionist paradigm': protected areas require strict protection; biodiversity conservation is a moral imperative; conservation linked to development does not protect biodiversity; harmonious, ecologically friendly local communities are myths; and emergency situations require extreme measures. We extend this analysis here to identify eight arguments supporting the move back to conservation strategies based on exclusive protected areas (Table 1).

These ideas have developed into a powerful global narrative, which we call 'back to the barriers'. This reasserts that biodiversity can only be conserved in areas free of all human influence (except science and limited ecotourism). It draws on conservation science to identify the best areas for reserves, and conceives of them as filling 'gaps' in a system or network of protected areas. It urges that all available resources are focused on expanding and completing this system, moving to 'landscape scale conservation' (da Fonseca *et al.*, 2005). It holds that such protected areas are best managed by centralised authorities, closely policing marked boundaries and applying appropriate sanctions on those who violate rules and borders, particularly those who enter reserves to reach the other side or to hunt, fish, gather plant materials or graze animals. Communities living around such reserves may be pacified by prominent investment in prominent social infrastructure (such as schools, roads or water supplies) and may even be engaged through some kind of local forum for information exchange, but they are excluded from decision-making processes about the management of protected area themselves or their place in the wider ecosystem and economy.

Table 1. Arguments supporting the ‘back to the barriers’ narrative¹

	<i>Argument</i>	<i>Explanation</i>	<i>References</i>
1	Biodiversity conservation is an overwhelming moral imperative.	Science shows the unprecedented scale and speed of biodiversity loss, and humanity has a moral duty to respond to this by preserving biodiversity.	Kramer and van Schaik, 1997; Terborgh, 1999
2	The movement towards community-based conservation represents an abandonment of clear scientific analysis in favour of ‘unscientific postmodernist influences’.	Science-based conservation priorities have been adversely affected by ‘unscientific postmodernist influences’. The notion that ideas about the state of species and ecosystems are socially constructed does not relate to ‘hard facts’ derived from ‘good science’, and undermines established priorities and arguments	Attwell and Cotterill, 2000
3	Community-based conservation is based on romantic and unrealistic ideas.	The community conservation approach drew part of its popularity from a romantic and inaccurate picture of ‘traditional’ communities and their resource management practices.	Redford, 1992; Redford et al., 1998; Redford and Mansour, 1996
4	So-called sustainable use is rarely if ever sustainable.	Weak common property institutions mean that harvesting of wild species is rarely if ever sustainable. CBNRM therefore does not stop the loss of biodiversity.	Hoyt, 1994; Ludwig, Hilborn and Walters, 1993
5	People-oriented approaches to biodiversity conservation have failed.	Wider community and development activities rarely achieve the survival of threatened species and ecosystems.	Oates, 1999; Wells and Brandon, 1992
6	Community approaches to conservation waste scarce conservation resources.	Community development and participation channel a significant portion of available funding away from biodiversity conservation yet produce minimal results in terms of biodiversity protection. Conservation should stop trying to deliver human benefits (development) at the local level and focus on the central goal of nature protection.	Oates, 1999
7	Protected areas have been proved to work.	National parks and other protected areas are necessary, effective and are the last safe havens for large tracts of tropical ecosystems. The need for protected areas is so great that all available funds should be devoted to their selection, establishment, protection and management.	Brandon, Redford and Sanderson, 1998
8	The scale of the threat to biodiversity demands the strictest possible protection of protected areas.	The dire threat to biodiversity demands extreme measures, including authoritarian enforcement practices.	Terborgh, 1999; Oates, 1999; Rabinowitz, 1999

¹ This table is a development of that in Wilshusen *et al.* (2002).

4. Actors and Factors Supporting the 'Back to the Barriers' Narrative

While the 'back to the barriers' narrative has been articulated most eloquently at an abstract level, by academic researchers in conservation biology, a number of other constituencies have fed ideas into the narrative or developed complementary activities that have served to support it. One constituency that appears to have had a pivotal impact is the bilateral and multilateral donor community, particularly the US Agency for International Development (USAID) and institutions linked to the World Bank. In the 1980s and early 1990s, donor agencies that help fund rural development and natural resource conservation in the developing world invested significant resources in community-based conservation and natural resource management schemes (Roe *et al.*, 2000). However, their focus swiftly changed in the late 1990s, coinciding with the rise of the 'back to the barriers' narrative. Detailed research to tease out relationships between donor funding priorities and the development of political narratives in conservation or development remains to be done, but in this case the coincidence is remarkable. Between 1996 and 1997, USAID in southern Africa dramatically switched its regional support from CBNRM to 'trans-boundary natural resource management' (TBNRM), after the publication of one of the first books raising concern about the role of social science and community considerations in conservation (Soulé and Lease, 1995), and shortly before the publication of a number of books critical of CBNRM (Brandon, Redford and Sanderson, 1998; Kramer, van Schaik and Johnson, 1997; Oates, 1999; Terborgh, 1999). The UK Department for International Development (DfID) phased out its support for CBNRM a little while later, though in this case it was not redirected elsewhere in the biodiversity resources sector. DfID subsequently withdrew direct support for biodiversity resource programmes in favour of support for 'sector reform' in recipient countries. In a statement to the UK Parliament on 1 April 2004, the Secretary of State for International Development, Hillary Benn, reported that while DfID was committed to working with the governments of Ghana and Cameroon to strengthen their capacity to deal with wildlife and poverty issues, it would do this by supporting sector reform. It was for the governments concerned to prioritise actions within their sector reform programmes.

The new enthusiasm of many donors has been for one or more of four sets of ideas, TBNRM, direct payments, public–private partnerships and mainstreaming. Their names reflect the way in which

donor agendas are built around keywords, which imply identity, novelty and a change from past unsuccessful approaches.

Trans-boundary natural resource management

The concept of trans-boundary conservation areas and ‘peace parks’ was introduced in the 1920s and by 1997 more than 100 such areas were to be found straddling 112 international borders in 98 countries (van der Linde *et al.*, 2001). In the mid-1970s the Peace Parks Foundation was formed in southern Africa and the regional office of USAID adopted TBNRM as its regional priority. It quickly ended its funding support for CBNRM activities, although some national USAID missions continued to support CBNRM, for example, in Namibia. The tendency of donor agencies to change their priorities, sometimes quite suddenly, and its influence on NGOs is well known (Edwards, Hulme and Wallace, 1999). The importance of donor funding to the success of different policy narratives (Roe, 1991) has been clearly illustrated in southern Africa where enthusiasm for TBNRM amongst government agencies, NGOs and consultants has been directly linked to the availability of donor funding (Duffy, 1997). The increased interest in TBNRM was in line with new landscape priority-setting exercises that were being developed by international conservation organisations and others, linked to the rise of systematic conservation planning (Margules and Pressey, 2000). The activities that were then generated, including academic research, fed back into the donor community to strengthen its support for TBNRM (Griffin *et al.*, 1999; Metcalfe, 1999; van der Linde *et al.*, 2001). While USAID remains one of the most high-profile supporters of TBNRM in southern Africa, the World Bank and other agencies implementing the Global Environment Facility (GEF) are also enthusiastic participants in the genre. Their support is based on Programme Element 1 of the Protected Areas Programme of Work on the Convention of Biological Diversity (CBD). This calls for the establishment and strengthening of trans-boundary protected areas (TBPAs) along with regional networks and collaboration between neighbouring protected areas across national boundaries (Goal 1.3).²

² The target for Goal 1.3 of the Protected Areas Programme of Work on the Convention on Biological Diversity (CBD) is to ‘establish and strengthen by 2010/2012 trans-boundary protected areas, other forms of collaboration between neighbouring protected areas across national boundaries and regional networks, to enhance the conservation and sustainable use of biological di-

TBNRM is envisaged as producing ecological, social and cultural, economic, financial, political and institutional opportunities for a wide range of actors, including rural communities and in many cases there is a strong rationale for the trans-boundary management of biodiversity (Cumming, 1999). However, while community engagement – usually of the sort that promises some sort of benefits rather than direct participation – is often central to the stated objectives of trans-boundary activities (see, for example, Lanjouw *et al.*, 2001) recent research from southern Africa suggests that most TBNRM initiatives are essentially based on the Peace Parks model in which the protected area agenda is foremost, and genuine support to community-based processes is not strong (Duffy, 1997; Wolmer, 2003). As it stands, although the TBNRM programmes that have replaced CBNRM may be intended to deliver a range of benefits in addition to biodiversity protection, their community dimensions are commonly overstated. With their central focus on protected areas, they are more accurately described simply as trans-boundary protected area programmes.

Direct payments

Another idea that has helped erode support for community-based activities is that of ‘direct payments’ for environmental services, including biodiversity conservation. This is based on growing research on the value of ecosystem services (see, for example, James *et al.*, 1999) and awareness that conservation imposes costs on local communities that needs to be paid by conservationists if conservation is to be equitable and effective (Balmford and Whitten, 2003). It also reflects the relative success of programmes in attempts to compensate farmers for profits foregone in managing their land for wildlife in Europe, and policy experiments with paying farmers in upland areas in Costa Rica for land management activities that deliver watershed protection (Miranda *et al.*, 2003). In the 1990s these ideas were extended to biodiversity conservation (Simpson and Sedjo, 1996) and by the start of the century were being adopted by donors who were suffering from CBNRM fatigue and were looking for new, more efficient approaches. A milestone in the introduction of direct payments in conservation occurred in 2002 at the 16th

versity, implementing the ecosystem approach, and improving international cooperation’. (www.biodiv.org/programmes/cross-cutting/protected/wopo.asp?prog=p1; 18 November 2004).

Annual Meeting of the Society for Conservation Biology held in Canterbury, UK, with a symposium entitled *Direct Payments as an Alternative Approach to Conservation Investment*, chaired by a staff member of the World Bank. The symposium organisers described a new ‘direct approach’ to conservation that would ‘supersede the fudges and failures of existing strategies for engaging with people living in biodiverse areas’. Their introduction to the symposium noted the frustration of donors, who had invested ‘billions of dollars’ in ‘indirect approaches’ that sought to protect ecosystems through targeted local development investments (ICDPs, CBNRM and similar approaches). It highlighted the ‘relatively new “direct approach”, in which conservationists identify priority areas for conservation and pay those who control these areas to protect the ecosystem from degradation. The payments are explicitly tied to conservation outcomes.’

The idea of direct payments has a great deal to commend it, and will doubtless be an important tool in conservation in the future, but it also raises a number of challenges, and threats. The principal problem is that direct payments can end up being an extension of conventional policies that exclude rural people from the resources they need for their livelihoods. Paying people to forgo opportunities to use species or ecosystems in particular ways may sound attractive, but it does little to build local management capacity or empower people to organise themselves to engage in broad environmental protection. The principle challenge is the technical one of establishing sustainable funding mechanisms. In the absence of both sustainable funding and organisational change at community-level direct payment schemes may not only be short term, but may result in the situation being worse at termination than at inception.

Public–private partnerships

The idea that cross-sectoral partnerships can be an important tool to release the economic potential of many developing countries is not new (Hamann and Acutt, 2003), and in South Africa there is a history of public–private partnerships in biodiversity conservation dating back to the 1980s when new national parks such as Pilanesberg were established (Farlam, 2005; Magome and Murombedzi, 2003). During the early days when this idea was being rolled out, the state tried, with mixed success, to use its assets such as protected areas to leverage business partnerships between rural communities and South Africa’s thriving private sector. More recently, the idea of pu-

blic–private partnerships for protected areas has begun to take root elsewhere in Africa – but the element of community involvement appears largely to have disappeared. The result is that this form of cross-sectoral partnership is emerging as a potential new threat to community-based conservation activity.

There is often widespread conflict between the interests of rural peoples and the interests of biodiversity conservation within protected areas. Time and time again the premise of many nature reserves across the developing world has been the same: the forcible uprooting of resident and mobile populations, often coerced violently to relocate somewhere else. In post-colonial East and southern Africa, involuntary relocation has taken place either during park creation programmes by the state, when the intended park areas are inhabited by poor indigenous populations, or during park ‘rehabilitation’ programmes by the state when people are ‘cleansed’ from legally-gazetted protected areas which they have occupied, or reoccupied, against government policy (see, for example, Brockington, 2002; Brockington and Schmidt-Soltan, 2004). The new enthusiasm for private investment in, and management of, nature reserves (notably in East and southern Africa) has the potential to be a new and potent force for social disruption in rural areas. These effects will have to be set against any biodiversity gains, and thus private conservation investment is an important political and policy issue.

Private investment in conservation has taken a variety of forms, including the management under contract of state-owned national parks by private individuals or companies; the incorporation of private land into state-owned parks; the establishment of private wildlife sanctuaries or hunting reserves; the creation of conservation trusts through land purchase by wealthy individuals or international NGOs; and the management by private companies under contract of community land set aside as a reserve. The degree of community involvement in and potential benefit from these schemes varies considerably, but all of them demonstrate the widespread acceptance of strictly protected areas as essential to conservation success. Many conservation projects based on private sector action are now receiving significant donor support, including from the GEF, and this is set to increase in the next five years or so. The draft Biodiversity Strategy submitted for discussion during the negotiations for the fourth replenishment of the GEF Trust Fund includes an action to ‘remove barriers to facilitate public–private partnerships’ (Global Environment Facility, 2005). Under this action, the GEF will support policy reform and/or incentives to catalyse engagement

of the private sector to attain improved financial sustainability of protected areas (PAs). GEF will also assist the private sector in the development of innovative ventures that demonstrate commercial profit and biodiversity benefit within the context of PAs.³

While private investment in conservation land will not necessarily result in involuntary relocation or the worsening of local poverty, it appears to have done so already on at least one occasion (Pearce, 2005) and in other cases there appears to be a direct conflict between the interests of private investors and local communities. Public-private partnerships for protected area management could have many advantages, but there is clearly a need for rigour in the design and implementation of such schemes, at the very least so that communities are not further disadvantaged. With some imagination it should surely be possible to emulate the early efforts by South Africa and be truly cross-sectoral, with meaningful community participation.

Mainstreaming

The most recent, and probably the most significant, prop to the 'back to the barriers' narrative is 'mainstreaming', which in the context of biodiversity is broadly equivalent to the call on Contracting Parties of the CBD to 'integrate...the conservation and sustainable use of biological diversity into relevant sectoral or cross-sectoral plans, programmes and policies' (Article 6(b)). The term 'mainstreaming' has been widely applied for many years with respect to issues such as gender (Beall, 1998), and used within the World Bank to describe the process through which global environmental objectives were taken into account in Country Assistance Strategies (Suter, J., personal communication); however, it was not mentioned in the CBD itself (agreed in 1992) and was almost totally absent from the GEF Corporate Business Plan for 2001–2004, presented to the 14th GEF Council Meeting in December 1999. It did appear in the 2003–2004 Business Plan (presented at the 19th Council Meeting in May 2002), but only in the context of the mainstreaming of GEF activities into the regular programmes of the GEF implementing agencies. In this plan, the strategic priorities for biodiversity were sustainable use activities within protected areas and their buffer zones, in support of biodiversity conservation, and 'conservation in productive landscapes and productive seascapes beyond formally

³ Strategic Objective One: Catalyzing Sustainability of Protected Area Systems at National Levels, Operational Activity d).

protected conservation areas and their buffer zones', recognising the scientific, ecological, and technical consensus that sustainable use was 'the only way to ensure long-term conservation'.

In the GEF's first decade, the emphasis in its biodiversity portfolio operational programmes had been on financing protected areas, with a smaller, although growing, engagement with sustainable use, mainstreaming and private-sector initiatives. The new strategy (for the GEF's second decade) explicitly stated that protected areas were the cornerstones of conservation, and proposed that biodiversity conservation be 'mainstreamed' by emphasising support for conservation beyond protected areas. The emphasis in such an approach would be on sustainability of results and the potential for replication, and move beyond a project-based approach to target the enabling environments in different countries and long-term institutional building systematically. The strategic priorities identified were, first, to catalyse sustainability of protected areas – to conserve biodiversity through the expansion, consolidation and rationalisation of national protected area systems – and, second, to mainstream biodiversity in production landscapes and sectors – that is, to integrate biodiversity conservation in agriculture, forestry, fisheries, tourism and other production systems and sectors.

The Biodiversity Strategy in the 2004–2006 Business Plan for the Third Replenishment (introduced to the 21st GEF Council Meeting in May 2003) established new strategic priorities. Sustainable use in protected areas and buffer zones had disappeared from amongst the strategic priorities, replaced by the concept of 'sustainability of protected areas', which is very different, and as a result the opportunities for support to CBNRM were reduced under the 2004–2006 Business Plan. This change reflects the way in which the 'back to the barriers' narrative pushed CBNRM aside in international thinking. We do not fully understand how this change took place. The changes in the third replenishment cannot have been a response to the CBD's Programme of Work on Protected Area, which was only adopted at the Seventh meeting of the Conference of Parties held in February 2004. However, the new emphasis on the 'sustainability of protected areas' does reflect text inserted into the Millennium Development Goals. These included protected areas (specifically 'ratio of area protected to maintain biological diversity to surface area') as indicator 26 for the achievement of MDG Goal 7 ('ensure environmental sustainability').⁴

⁴ http://unstats.un.org/unsd/mi/mi_goals.asp (18 November 2005).

In May 2003, the language of the second strategic priority in the 2004–2006 Business Plan was changed to introduce the concept of ‘mainstreaming’. This might appear to encourage community-based approaches to conservation around and away from protected areas; however, it has to be understood in the full context of the changes that took place in the third replenishment. Once the Business Plan had been accepted the GEF Members, Council and Secretariat, its implementing agencies, national biodiversity agencies and NGOs found themselves within a policy environment in which protected areas had become a priority for GEF funding, arguably the most influential funding agency in this arena. Sustainable use strategies apply anywhere in the landscape (including protected areas). However, the idea of mainstreaming focuses explicitly on land outside protected areas. This effectively created a conceptual separation between protected areas and their surroundings, allowing separate funding streams for protected areas and the wider landscape. The new Business Plan therefore created a stable policy environment for the ‘back to the barriers’ agenda, since mainstreaming in the wider landscape could coexist comfortably with the support of preservation in protected areas, in a classically compartmentalised landscape.

There is still some scope for community-based conservation based on the sustainable use of biodiversity, but the implementation of mainstreaming in the fourth replenishment moves concern away from community to sectoral level. The stated aim is ‘to internalize the goals of biodiversity conservation and its sustainable use into production systems, supply chains, markets, sectors, development models, policies and programs’. An important element of this work is the encouragement of more private-sector business actors. Admireable although this is, it does not encourage direct support for community-level sustainable management of resources, or community conservation initiatives. Overall, the GEF Biodiversity Strategy provides little encouragement to CBNRM, but sustains the ‘back to the barriers’ narrative.

5. The Limits of Barrier Approaches

We are not arguing that protected areas, even strict ‘no entry’ PAs, have no place in conservation strategies. However, we feel that there is a real risk that a focus on exclusion will lead to conservation strategies that so lack democratic legitimacy that they will not endure. PAs are not a ‘magic bullet’ to deal with biodiversity loss.

The eight arguments for a move ‘back to barriers’ discussed earlier (Table 1) are important, but they are also flawed. Wilshusen *et al.* (2002) pick up on some of these flaws. We develop their analysis here (Table 2), simply to indicate the importance of not getting carried away by the appeal of the ‘back to barriers’ narrative.

Table 2. Critique of the ‘back to the barriers’ narrative ⁵		
Argument		Critique
1 Biodiversity conservation is an overwhelming moral imperative.	a)	Accepting the moral argument for biodiversity conservation does not mean that it should take priority over the separate moral claims of people with rights to land and resources.
	b)	Priority given to scientific strategies for protecting biodiversity can limit dialogue with other groups, including local communities, and can constrain efforts to identify creative solutions that carry legitimacy for all parties.
	c)	It is wrong to consider that local interests should never supersede national and global interest in biodiversity. Local people carry a disproportionate burden in terms of negative social, cultural and economic impacts.
2 The movement towards community-based conservation represents an abandonment of clear scientific analysis in favour of ‘unscientific postmodernist influences’.	a)	Conservation is and always has been about the choices that human society makes. Science can inform such choices, but not determine them.
	b)	Science is not the only way of understanding and appreciating nature. Even scientists respond to species and ecosystems in complex ways that stretch beyond the narrow dictates of their science.
	c)	Science works by disproving hypothesis; at no point in time can it offer certainty.
	d)	Science offers no ‘short-cut’ around the difficult decisions that society has to take if it is to conserve biodiversity.

⁵ This table is a development of that in Wilshusen *et al.* (2002).

3 Community-based conservation is based on romantic and unrealistic ideas.	a) Some conservation literature does tend to glorify indigenous peoples specifically, and 'traditional' communities more generally, but it is important not to ignore the evidence that in many instances such communities have demonstrated their capacity to manage natural resources effectively over time. b) Critics over-generalise in the opposite direction, creating the false impression that all resource-dependent peoples act only to maximise short-term gains without considering wider political and economic factors that might encourage them to do so. c) Sceptics of community management tend to ignore decision making, organisational and governance processes – both customary and modern – that structure resource use within and among rural communities. d) Critics underestimate the possibility that, even in the face of rapid change, traditional groups might be able to adapt patterns of use to a more sustainable pathway, especially with outside support.
4 Sustainable use is rarely if ever truly sustainable in terms of biological impact.	a) While extractive use of wild species has often been unsustainable, this has as much, if not more to do with weak common property institutions than biology. b) The common property literature shows the conditions necessary for effective management institutions. Promoting these is challenging but not impossible. c) Poor people have a real incentive to maintain species and habitats if they generate economic benefits to which they have entitlement. There is no incentive to maintain species from which there is no realistic prospect of future benefit. In practice, benefits from illegal hunting are typically not balanced against likely future benefits from sustained harvesting, but against risk of capture and punishment. Patterns of incentives may lead to perverse outcomes, but such incentives can be changed by economic, political and social processes. d) The sustainable use of ecosystems by communities may be the only viable political and economic strategy for broad landscape conservation in many countries.

5 People-oriented approaches to biodiversity conservation have failed.	a) The perceived lack of effectiveness may well be real in many instances, but this usually has more to do with the way programmes are implemented rather than the concept. b) Biodiversity conservationists have commonly paid lip service to community-based approaches, only adopting them as a secondary strategy to achieve their primary objective when their 'preferred' strategy of protection has limited options or has failed. Under these conditions it is unsurprising that CBNRM does not work either. c) There are clear cases where communities succeed in managing resources, even in the face of global pressures such as resource commercialisation. d) Most criticism related to 'effectiveness' overlooks or discounts important processes such as organisational strengthening that have emerged from many projects.
6 Community approaches to conservation waste scarce conservation resources.	a) While significant international donor funding was invested in community-based approaches in the 1980s and 1990s, the bulk of these funds were from rural development budget-lines. This represents new money for conservation. Had they not been spent on CBNRM it is highly unlikely that they would have been available for conservation purposes. b) It is likely that donor funding had the effect of expanding conservation activities into the agricultural landscape, rather than reducing conservation funding for protected areas. c) Conservation alone may not be able to bear the burden of development, but it cannot simply disengage from the social and political context by leaving the 'social work' to others. d) Abandoning community approaches in favour of protectionism alone will waste social capital created by existing attempts to build alliances, strengthen organisations, and negotiate programs with people and governments in developing countries.

7 Protected areas have been proved to work.	a) Protected areas often do not work well. Forty per cent of tropical protected areas are in no better condition than the land outside with respect to human activities such as hunting and grazing (Bruner et al., 2001). b) Most protected areas in biodiversity-rich tropical countries are significantly underfunded, which limits their effectiveness. Effectiveness is rarely and poorly measured. c) Corruption is a real constraint on the effectiveness of protected areas. Such impacts are rarely acknowledged or measured. d) Scientific understanding of protected area management has advanced, but the greatest challenges to protected areas are political and economic. These are less well understood and dealt with. e) Rural communities often associate protected areas with the control of land by an elite or rival group, and attribute wider local failings of governance to the way conservation is implemented. This breeds opposition to conservation. f) Communities, activists, government officials and others in developing countries commonly feel that the emphasis on protected areas by international conservation organisations is a manifestation of external control that mirrors past imperial domination. These issues are rarely explicitly discussed. g) The negative social impacts of protected areas (e.g. evictions and exclusions) can provoke popular resistance to conservation at a level that could potentially offer a political challenge to democratic governments. h) Fair and legitimate enforcement of protected area regulations is not impossible but it is more costly and time-consuming than is commonly realised. The short time-frames of most project appraisals do not allow proper assessment of conservation success: the success of protected areas may be short-lived and dependent on levels of outside funding that cannot be sustained.
--	--

8 The scale of the threat to biodiversity demands the strictest possible protection of protected areas.	a) In addition to being ecologically sound, conservation interventions must be socially and politically feasible and morally just. If not, they will generate resistance and conflict, derailing attempts at protection. b) Approaches to the biodiversity crisis that involve emergency enforcement measures to guarantee species and habitat protection may produce short-term gains, but undermine possibilities for biodiversity protection over time. Advocates of strict protection underestimate the cost and practicality of strict protection, and overestimate local views of the legitimacy of governments and other actors who lead such approaches. c) There is a tendency for governments using military personnel and methods in conservation to combine this with other strategic or political goals in a way that threatens the legitimacy and sustainability of conservation outcomes. d) National level enforcement is most effective where it compliments local institutions' moral economy. Voluntary compliance associated with bottom-up processes can support effective enforcement. e) Strong militaristic enforcement to achieve conservation outcomes would be best deployed to support the interests of the poor, for example to secure local rights that underpin resource use at sustainable levels in the face of outside actors (e.g. poachers).
--	---

This suite of counter-arguments does, we feel, significantly dent the case for abandoning community-based approaches to conservation in favour of an exclusive, preservationist strategy. Protected areas are obviously important for conservation, and have a role in a wider economic and cultural landscape, but are unsatisfactory as a sole focus of conservation strategy.

6. Is there Life in CBNRM?

An important argument of those who argue for a renewed focus on exclusive, protected areas is that community approaches, particularly CBNRM, have failed. It is ironic that much of the information that has fuelled this critique has been produced by advocates of

community approaches themselves. Self-critical analysis, aimed at improving the effectiveness and reach of conservation approaches centred around human needs and the incentives produced by the convergence of devolved authority, accountability and responsibility, has provided fuel for the rejection of the community approach as a whole, particularly that part of the approach based on CBNRM.

It is certainly the case that there is now a rich literature, from southern Africa in particular, which analyses the shortcomings of community-based approaches to conservation. CBNRM programmes in southern Africa have been subject to extensive analysis and, while the early evaluations of the programmes generally consisted of glib praise of the potential benefits of devolving control and authority over natural resources to communities (see, for example, Metcalfe 1994), the evaluations – many, if not most, of which came from academics and practitioners involved in these programmes – became progressively more sophisticated, nuanced and negative (Duffy, 2000; Dzingirai, 1995; Gibson and Marks, 1995; Hill, 1996; Hughes, 2001; Jones, 2001; Katerere, 2002; Murombedzi, 1999, 2001; Murphree, 1994, 2001; Wainwright and Wehrmeyer, 1998).

The most widely shared conclusion appears to be that CBNRM in this region is indeed flawed, but not because communities are inherently unable to control themselves or their resources. Most analysts and practitioners alike agree that the policy and legislative reforms have not resulted in sufficient community control over natural resources. Typically, there has been no real devolution of power and authority over resources, including land, from the state to local people (Murombedzi, 2001). At best these have been devolved from central government to local authorities or some other level of government, and as such CBNRM has resulted in insufficient incentives for communities to internalise the costs of resource management (Murphree, 2001; Jones, 2001). Decentralisation *per se* is not adequate to create the conditions required for significant community control over natural resources (Ribot and Larson, 2004). Scholars and practitioners in the field have suggested that what is actually required is a process that gives communities control over their resources through tenure reform (Murombedzi, 2001). A more radical view of CBNRM holds that, rather than being designed to benefit the rural poor living under communal tenure, CBNRM has actually perpetuated the unresolved coexistence of dual tenure systems, with communal tenure continuing to function in ways that disadvantage tenants relative to those enjoying freehold and leasehold rights. On top of this CBNRM has offered opportunities

for the post-colonial state to extend its influence to areas that had hitherto been beyond its reach, largely due to resource limitations (Hill, 1996). A complementary argument holds that CBNRM has often been weakened by its failure to engage with conventional rural development policy constituencies in either agriculture or land reform (Murombedzi, 2001).

The reasons for the poor performance of CBNRM projects therefore range from the poor quality of project design and the unqualified nature of many of those attempting implementation, to major policy failure in the devolution of power and authority (Dzingirai, 1995; Gibson and Marks, 1995; Hill, 1996; Hughes, 2001; Jones, 2001; Katerere, 2002; Murombedzi, 1999, 2001; Murphree, 1994, 2001; Wainwright and Wehrmeyer, 1998). However, instead of using these shortcomings as an excuse to abandon community approaches – especially in favour of a strategy which, itself, has a questionable record of success – we agree with Murphree (2000) when he asserts that community-based conservation has ‘not been tried and found wanting: it has been found difficult and rarely tried’. The challenge is not how best to replace community approaches, but how to design and implement them so that they reach their potential. After all, even if it was agreed that the intrinsic rights of nature supersede those of people, the ‘protection-by-any-means-necessary’ approach is likely to fail on pragmatic grounds.

7. Conclusions

Biodiversity is increasingly under threat, but its conservation is not easily achieved. Amongst the suite of mechanisms available to tackle the problem, current approaches, formulated principally by biodiversity scientists and international NGOs, tend to support the primacy of areas of land that are protected against human activity. We believe that protected areas have had, and will continue to have, value for biodiversity conservation, ecosystem services as well as broader human welfare. We do not dispute that protected areas can be legitimate and effective, nor that strict human exclusion may be appropriate and possible in some cases. However, we also recognise the reality that both the effectiveness and sustainability of protected areas is often under considerable doubt, as witnessed by the fact that 40 per cent of tropical parks are in many ways in no better condition than the land outside them (Bruner *et al.*, 2001). As Brechin *et al.* (2002) point out, ‘most areas considered to be high priority biodiversity “hotspots” are also social and political

“hotbeds”, which makes the challenge of doing conservation very complex. Where conservation programmes tie up natural resources sought after by resource-dependent agrarian communities, locally driven processes are the biggest threat to protected areas, a situation that will be exacerbated if genuine democracy takes root and local people gain political power. In these conditions, science-led solutions alone will not be enough to safeguard biodiversity. Nor will authoritarian approaches. Establishing legitimate processes by constructively working with people will be the most feasible and morally just way to achieve long-term nature protection (Brechtin *et al.*, 2002), and this will inevitably result in programmes of sustainable use that revolve around communities – because the status quo in much of the developing world means that land and biodiversity resources can only be managed above the level of the individual household.

A further problem with fortress conservation and the ‘back to the barriers’ narrative is that, even if they all worked, protected areas cannot do the whole job. Not only are they becoming increasingly isolated, which threatens the long-term conservation of many species (DeFries *et al.*, 2005) but we need effective conservation activity in the other 90 per cent of the landscape where people make their livelihoods and where sustainable use and community-based approaches may be amongst the few practical options (Hutton and Leader-Williams, 2003). Mainstreaming undoubtedly recognises the importance of conservation outside protected areas, but whether this will develop as an effective conservation tool on the ground remains to be seen. At the moment, because it is predicated on the idea that protected areas are central to biodiversity conservation, and because it is not focusing on community-based approaches to conservation (at least in the context of the GEF), mainstreaming supports and will comfortably coexist with fortress conservation. Indeed, it may have the result of entrenching this narrative for the foreseeable future and in the absence of other important processes we could be seeing the ‘end of history’ (Fukuyama, 1992) so far as the political narratives in biodiversity conservation are concerned. However, other processes do exist. In particular the international consensus regarding poverty alleviation has already started to call into question biodiversity policy and practice (see, for example, Adams *et al.*, 2004) and it would be hasty to predict what the discourse around the conservation of biodiversity resources might look like in a decade.

TBNRM, direct payments and public–private partnerships sup-

port the 'back to the barriers' narrative, and they undoubtedly have a role to play in securing biodiversity. However, it seems likely that in many, if not most instances they will have to be deployed in conjunction with or as part of broader community-based activities if they are to achieve their potential. In suggesting this we do not mean to give the impression that we are uncritically supportive of community-based approaches to conservation. We readily accept that skepticism towards community-based approaches is both healthy and very often well placed. It should, however, be part of a learning process, not a campaign for extinction, and in this regard strident calls for community-based approaches to be abandoned are both premature and irresponsible. The issue for us is not whether there should be biodiversity conservation, but how this is best achieved. It is not about protected areas, but how they are implemented. Similarly, it should not be about community-based approaches, but how they too are implemented.

In southern Africa, which for many people has been the spiritual home of CBNRM, there have now been almost two decades of experience of the approach. Given the developments we have described, it is important to focus on the lessons learned and to embark on a thorough discussion so that a way can be charted forward through the shortcomings, disappointments and failures. If southern Africa cannot demonstrate that community-based approaches to biodiversity and natural resource management can deliver effectively, then the only alternative will indeed be a return to conservation only behind hard 'barriers'. We have no confidence that such an exclusive strategy will lead to outcomes satisfactory in either human or ecological terms.

References

- Adams, N.S. and T.O. McShane, 1992, *The Myth of Wild Africa: Conservation without Illusion*, New York: W. W. Norton.
- Adams, W.M., 2003, 'Nature and the colonial mind', pp. 16–50 in Adams and Mulligan, eds.
- Adams, W.M., 2004, *Against Extinction: the Story of Conservation*, London: Earthscan.
- Adams, W.M. and D. Hulme, 2001, 'Conservation and Communities: Changing Narratives, Policies and Practices in African Conservation', pp. 9–23 in Hulme and Murphree, eds.
- Adams, W.M. and M. Mulligan, eds, 2003, *Decolonizing Nature: strategies for conservation in a post-colonial era*, London: Earthscan.
- Adams, W.M., R. Aveling, D. Brockington, D. Dickson, J. Elliot, J. Hutton, B. Vira and W. Wolmer, 2004, 'Conservation and Poverty: A framework for Analysis',

- Science*, Vol. 306, pp. 1146–1148.
- Attwell, C.A.M. and F.P.D. Cotterill, 2000, 'Postmodernism and African Conservation Science', *Biodiversity Conservation*, Vol. 9, pp. 559–577.
- Balmford, A. and T. Whitten, 2003, 'Who should pay for tropical conservation, and how could the costs be met?' *Oryx*, Vol. 37, pp. 238–250.
- Barrett, C.B. and P. Arcese, 1995, 'Are Integrated Conservation-Development Projects (ICDPs) Sustainable? On the conservation of large mammals in sub-Saharan Africa', *World Development*, Vol. 23, pp. 1073–1084.
- Barrow, E. and M. Murphree, 2001, 'Community conservation: from concept to practice', pp. 24–37 in Hulme and Murphree, eds.
- Beall, J., 1998, 'Trickle-down or Rising Tide? Lessons on Mainstreaming Gender Policy from Colombia and South Africa', *Social Policy and Administration*, Vol. 32, No. 5, pp. 513–518.
- Benjaminsen, T.A., B. Cousins and L. Thompson, eds, 2002, *Contested Resources: Challenges to the Governance of Natural Resources in Southern Africa*, Cape Town: Programme for Land and Agrarian Studies.
- Brandon, K., K.H. Redford and S.E. Sanderson, eds, 1998, *Parks in Peril: People, Politics and Protected Areas*, Washington DC: Island Press.
- Brechin, S.R., P.R. Wilshusen, C.L. Fortwangler and P.C. West, 2002, 'Beyond the Square Wheel: Toward a More Comprehensive Understanding of Biodiversity Conservation as Social and Political Process', *Society and Natural Resources*, Vol. 15, pp. 41–64.
- Brechin S.R., P.R. Wilshusen, C.L. Fortwangler, eds, 2003, *Contested Nature: promoting International Biodiversity with Social Justice in the Twenty-first Century*, New York: State University of New York Press.
- Brockington, D., 2002, *Fortress Conservation: the preservation of the Mkomazi Game Reserve, Tanzania*, Oxford: James Currey.
- Brockington, D. and K. Schmidt-Soltau, 2004, 'The social and environmental impacts of wilderness and development', *Oryx*, Vol. 38, pp. 140–142.
- Brosius, J.P., 1999, 'Analyses and interventions: anthropological engagements with environmentalism', *Current Anthropology*, Vol. 40, pp. 277–309.
- Bruner, A.G., R.E. Gullison, R. E. Rice and G.A.B. da Fonseca, 2001, 'Effectiveness of Parks in Protecting Tropical Biodiversity', *Science*, Vol. 291 (5501), pp. 125–128.
- Carruthers, J., 1995, *The Kruger National Park: a social and political history*, Pietermaritzburg: University of Natal Press.
- Chapin, M., 2004, 'A Challenge to conservationists', *Worldwatch Magazine*, November/December 2004.
- Child, B., ed., 2004, *Parks in Transition: Biodiversity, rural development and the bottom line*, London: Earthscan.
- Colchester, M., 1997, 'Salvaging nature: indigenous peoples and protected areas', pp. 97–130 in Ghimire and Pimbert, eds.
- Colchester, M., 1998, 'Who will garrison the fortress? A reply to Spinage', *Oryx*, Vol. 32, pp. 245–248.
- Colchester, M., 2003, *Salvaging nature: indigenous peoples, protected areas and biodiversity conservation*, Montevideo: World Rainforest Movement.
- Cumming, D.H.M., 1999, 'Study on the Development of Transboundary Natural Resource Management Areas in Southern Africa – Environmental Context:

- Natural Resources, Land Use, and Conservation', Washington DC: Biodiversity Support Program.
- da Fonseca, G.A.D., A. Bruner, R.A. Mittermeier, K. Alger, C. Gascon and R.E. Rice, 2005, 'On defying nature's end: the case for landscape-scale conservation', *The George Wright Forum*, Vol. 22, No. 1, pp. 46–60.
- DeFries, R., R. Hansen, A.C. Newton and M.C. Hansen, 2005, 'Increasing isolation of protected areas in tropical forests over the past twenty years', *Ecological Applications*, Vol. 15, No. 1, pp. 19–26.
- Duffy, R., 1997, 'The Environmental Challenge to the Nation-State: Superparks and National Parks Policy in Zimbabwe', *Journal of Southern African Studies*, Vol. 23, No.3, pp. 441–451.
- Duffy, R., 2000, *Killing for Conservation: wildlife policy in Zimbabwe*, Oxford: James Currey.
- Dzingirai, V., 1995, *Take back your CAMPFIRE: a study of local level perceptions to electric fencing in the framework of Binga's CAMPFIRE programme*, Mount Pleasant, Harare: Centre for Applied Social Sciences, University of Zimbabwe, 1995 (CASS Occasional Paper Series – NRM).
- Edwards, M., D. Hulme and T. Wallace, 1999, 'NGOs in a global future: marrying local delivery to worldwide leverage', *Public Administration and Development*, Vol. 19, pp. 117–136.
- Fabricius, C., E. Koch, H. Magome and S. Turner, eds, 2004, *Rights, Resources and Rural Development: community-based natural resource management in southern Africa*, London: Earthscan.
- Farlam, P., 2005, *Working Together Assessing Public–Private Partnerships in Africa*, South African Institute of International Affairs, Nepad Policy Focus Report No. 2, 59 pp.
- Ferraro, P.J. and R.D. Simpson, 2000, *The Cost-Effectiveness of Conservation Payments*, Discussion Paper 00–31, Resources for the Future, 1616 P Street, NW, Washington DC, 20036.
- Fukuyama, F., 1992, *The End of History and the Last Man*, Harmondsworth: Penguin Books.
- Ghimire, K. and M. Pimbert, eds, 1997, *Social Change and Conservation: environmental politics and impacts of national parks and protected areas*, London: Earthscan.
- Gibson, C.C. and S.A. Marks, 1995, 'Transforming rural hunters into conservationists: an assessment of community-based wildlife management programs in Africa', *World Development*, Vol. 23, pp. 941–957.
- Global Environment Facility, 2005, *Working draft GEF Biodiversity Strategy*, Doc GEF/R.4/Inf.6 August 10, 2005, Meeting on the Fourth Replenishment of the GEF Trust Fund September 2, 2005 Washington DC.
- Griffin, J., D. Cumming, S. Metcalfe, M. t'Sas-Rolfes, J. Singh, E. Chonguica, M. Rowen and J. Oglethorpe, 1999, *Study on the Development of Transboundary Natural Resource Management Areas in Southern Africa*, Washington DC: Biodiversity Support Program.
- Grove, R.H., 1995, *Green Imperialism: colonial expansion, tropical island Edens and the origins of environmentalism, 1600–1800*, Cambridge: Cambridge University Press.
- Hamann, R. and N. Acutt, 2003, 'How should civil society (and the government) respond to "corporate social responsibility"? A critique of business motiva-

- tions and the potential for partnerships', *Development Southern Africa*, Vol. 20, No. 2, pp. 255–270.
- Hannah, L., 1992, *African People, African Parks*, Washington DC: Conservation International.
- Hill, K. A., 1996, 'Zimbabwe's CAMPFIRE Program: Grassroots Democracy or Government Co-optation?' *African Studies Review*, Vol. 39, No. 1, pp. 103–121, April.
- Hoyt, J., 1994, *Animals in Peril: How sustainable use is wiping out the world's wildlife*, New York: Avery Publishing Group.
- Hughes, D.M., 2001, 'Rezoned for business: how ecotourism unlocked black farmland in eastern Zimbabwe', *Journal of Agrarian Change*, Vol. 1, pp. 575–599.
- Hulme, D. and M. Murphree, 2001, eds, *African Wildlife and Livelihoods: the promise and performance of community conservation*, Oxford: James Currey.
- Hutton, J.M. and N. Leader-Williams, 2003, 'Sustainable use and incentive-driven conservation: realigning human and conservation interests', *Oryx*, Vol. 37, pp. 215–226.
- James, A.N., K.J. Gaston and A. Balmford, 1999, 'Balancing the earth's accounts', *Nature*, Vol. 401, pp. 323–324.
- Jones, B., 2001, 'The evolution of community-based approaches to wildlife management in Kunene, Namibia', pp. 160–176 in Hulme and Murphree, eds.
- Katerere, Y., 2002, 'Community-public-private Partnerships in CBNRM: The Real Challenges?' pp. 20–39 in Benjaminsen, Cousins and Thompson, eds.
- Kemf, E., 1993, ed., *The Law of the Mother: protecting indigenous peoples in protected areas*, San Francisco: Sierra Club Books.
- Kramer, R.A., C.P. van Schaik and J. Johnson, eds, 1977, *Last Stand: Protected areas and the defense of tropical biodiversity*, New York: Oxford University Press.
- Lanjouw, A., A. Kayitare, H. Rainer, E. Rutagarama, M. Sivha, S. Asuma and J. Kalpers, 2001, *Transboundary Natural Resource Management: A Case Study by International Gorilla Conservation Programme in the Virunga-Bwindi Region*, Washington DC: Biodiversity Support Program.
- Leach, M. and R. Mearns, eds, 1996, *The Lie of the Land: challenging received wisdom in African environmental change*, Oxford: James Currey.
- Ludwig, D., R. Hilborn and C. Walters, 1993, 'Uncertainty, Resource Exploitation and Conservation: Lessons from History', *Science*, Vol. 260, p. 17.
- MacKenzie, J. M., 1988, *The Empire of Nature: Hunting, Conservation and British Imperialism*, Manchester: Manchester University Press.
- McNeely, J.A. and K.R., Miller, eds, 1984, *National Parks, Conservation and Development: the role of protected areas in sustaining society*, Washington DC: Smithsonian Institution Press.
- McNeely, J., ed., 1992, *Parks for Life: report of the IVth World Congress on National Parks and Protected Areas*, Gland, Switzerland: IUCN.
- Magome, H. and J. Murombedzi, 2003, 'Sharing South African national parks: community land and conservation in a democratic South Africa', pp. 108–134 in Adams and Mulligan, eds.
- Margules, C.R. and R.L. Pressey, 2000, 'Systematic conservation planning', *Nature*, Vol. 405, pp. 243–253.

- Metcalfe, S., 1994, 'The Zimbabwe Communal Areas Management Programme for Indigenous Resources (CAMPFIRE)', pp. 161–192 in Western, Wright and Strum, eds.
- Metcalfe, S.C., 1999, *Study on the Development of Transboundary Natural Resource Management Areas in Southern Africa – Community Perspectives*, Washington, DC: Biodiversity Support Program.
- Miranda, M., I.T. Porras and M.L. Moreno, 2003, *The social impacts of payments for environmental services in Costa Rica: A quantitative field survey and analysis of the Virilla watershed*, London, UK: IIED publications.
- Murombedzi, J.S., 1999, 'Devolution and stewardship in Zimbabwe's CAMPFIRE Programme', *Journal of International Development*, Vol. 11, pp. 287–293.
- Murombedzi, J., 2001, 'Why wildlife conservation has not economically benefited communities in Africa', pp. 208–226 in Hulme and Murphree, eds.
- Murphree, M., 1994, 'The role of institutions in community-based conservation', pp. 403–427 in Western, White and Strum, eds.
- Murphree, M., 2000, 'Community-based Conservation: Old Ways, New Myths and Enduring Challenges', key address at the conference on African Wildlife Management in the New Millennium, 13–15 December, Mweka, Tanzania: College of African Wildlife Management.
- Murphree, M., 2001, 'A case study of ecotourism development from Mahenye, Zimbabwe', pp. 177–194 in Hulme and Murphree, eds.
- Myers, N., R.A. Mittermeier, C.G. Mittermeier, G.A.B. de Fonseca and J. Kent, 2000, 'Biodiversity hotspots for conservation priorities', *Nature*, Vol. 403, pp. 853–858.
- Neumann, R.P., 1998, *Imposing Wilderness: struggles over livelihood and nature preservation in Africa*, Berkeley, CA: University of California Press.
- Oates, J.F., 1995, 'The dangers of conservation by rural development: A case study from the forests of Nigeria', *Oryx*, Vol. 29, pp. 115–122.
- Oates, J.F., 1999, *Myth and reality in the rainforest: How conservation strategies are failing in West Africa*, Berkeley: University of California Press.
- Pearce, F., 2005, 'Humans losing out to Africa's big game industry', *New Scientist*, 16 April, No. 2495, p. 21.
- Pierce, S.M., R.M. Cowling, T. Sandwith and K. MacKinnon, 2002, *Mainstreaming Biodiversity in Development*, Washington DC: World Bank.
- Rabinowitz, A., 1999, 'Nature's last bastions: Sustainable use of our tropical forests may be little more than wishful thinking', *Natural History*, Vol. 108, pp. 70–72.
- Redford, K.H., 1991, 'The ecologically noble savage', *Cultural Survival Quarterly*, Vol. 15, pp. 46–48.
- Redford, K.H., 1992, 'The empty forest', *Bioscience*, Vol. 42, pp. 412–422.
- Redford, K.H. and J. Mansour, eds, 1996, *Traditional peoples and biodiversity conservation in large tropical landscapes*, Rosslyn, VA: The Nature Conservancy.
- Redford, K.H., K. Brandon and S.E. Sanderson, 1998, 'Holding Ground', pp. 455–464 in Brandon, Redford and Sanderson, eds.
- Ribot, J. and A. Larson, 2004, eds, *Decentralization of Natural Resources: Experiences in Africa, Asia and Latin America*, London: Routledge.
- Roe, D., J. Mayers, M. Grieg-Gran, A. Kothari, C. Fabricius and R. Hughes,

- 2000, *Evaluating Eden: Exploring the myths and realities of community-based wildlife management: Series overview*, London: IIED Evaluating Eden, IIED Publications.
- Roe, E., 1991, 'Development narratives, or making the best of blueprint development', *World Development*, Vol. 19, pp. 287–300.
- Shafer, C.L., 1990, *Island theory and conservation practice*, Washington: Smithsonian Institution Press.
- Simpson, R.D. and R.A. Sedjo, 1996, 'Paying for the Conservation of Endangered Ecosystems: A Comparison of Direct and Indirect Approaches', *Environment and Development Economics*, Vol. 1, pp. 241–257.
- Smith, R.J., R. D.J. Muir, M.J. Walpole, A. Balmford and N. Leader-Williams, 2003, 'Governance and the loss of biodiversity', *Nature*, Vol. 426, pp. 67–70.
- Soulé, M. E., 1995, 'The social siege of nature', pp. 37–170 in Soulé and Lease, eds.
- Soulé, M.E. and G. Lease, eds, 1995, *Reinventing Nature? Responses to postmodern deconstruction*, Washington: Island Press.
- Steiner, A., 2003, 'Trouble in paradise', *New Scientist*, Vol. 180 (2417), p. 21.
- Stocking, M. and S. Perkin, 1992, 'Conservation-with-development: an application of the concept in the Usambara Mountains, Tanzania', *Transactions of the Institute of British Geographers N.S.*, Vol. 17, pp. 337–349.
- Terborgh, J., 1999, *Requiem for Nature*, Washington DC: Island Press/Shearwater Books.
- Turner, M. and D. Hulme, 1997, *Governance, Administration and Development*, London: Macmillan.
- van der Linde, H., J. Oglethorpe, T. Sandwith, D. Snelson and Y. Tessema (with contributions from A. Tiéga and T. Price), 2001, *Beyond Boundaries: Transboundary Natural Resource Management in Sub-Saharan Africa*, Washington DC: Biodiversity Support Program.
- Wainwright, C. and W. Wehrmeyer, 1998, 'Success in integrating conservation and development? A study from Zambia', *World Development*, Vol. 26, pp. 933–44.
- Wells, M. and K. Brandon, 1992, *People and Parks: Linking Protected Areas with Local Communities*, Washington DC: World Bank.
- Western, D., M. Wright and S. Strum, eds, 1994, *Natural Connections: Perspectives in Community-based Conservation*, Washington DC: Island Press.
- Wilshusen, P.R., S.R. Brechin, C. L. Fortwangler and P. C. West, 2002, 'Reinventing a Square Wheel: Critique of a Resurgent "Protection Paradigm" in International Biodiversity Conservation', *Society and Natural Resources*, Vol. 15, pp. 17–40.
- Wilson, E.O., 1992, *The Diversity of Life*, Cambridge, MA: Harvard University Press.
- Wolmer, W., 2003, 'Transboundary Protected Area governance: tensions and paradoxes', paper prepared for the workshop on Transboundary Protected Areas in the Governance Stream of the 5th World Parks Congress, Durban, South Africa.